



One Lake, Two Basins – Does Productivity Define Fish and Food Web Mercury Dynamics in Human-impacted Lake Vesijärvi, Southern Finland?

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Received: 27 May 2026 / Accepted: 21 August 2026 / Published online: 2 September 2026
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Abstract Productivity in large lakes is often heterogeneous between basins, with potential effects on the fish community and contamination levels in the food web. Clean lakes and resilient food webs support biodiversity and ecological integrity, however, pressures on these ecosystems are numerous. Mercury (Hg) is a ubiquitous neurotoxic metal linked to productivity, thus highlighting its potential value for lake monitoring. However, limited knowledge exists regarding the effectiveness of Hg monitoring between basins of large lakes with different nutrient concentrations. In this study, primary producers, benthic macroinvertebrates, zooplankton, and the fish community were sampled in the less productive Kajaanselkä and more productive Enonselkä basins of Lake Vesijärvi, southern Finland. Basin differences in total Hg content [THg] (mg kg^{-1}) and its bioaccumulation in common fish species, and biomagnification in the

food web were assessed. In fish, [THg] was generally higher in the Kajaanselkä basin, although not in all species. Bioaccumulation regression slopes between [THg] and length were also not consistently different between basins. Biomagnification was detected in both basins through the regression of [THg] with nitrogen isotopes ($\delta^{15}\text{N}$). However, the trophic magnification slopes (TMS) and [THg] baselines (intercepts) were similar. Variation in [THg] and bioaccumulation found in some fish likely indicates differences in species-specific habitat and resource use and availability between basins. These methods revealed that basin differences in fish THg metrics did not translate to differences in food web biomagnification between basins, although the methods did highlight the potential monitoring value of pelagic fish in large boreal lakes. Similar methods may give more pronounced results given larger differences in morphology and physical and chemical characteristics between lake basins, and future work may

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s11270-026-09890-7>.

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explore basin-specific differences in biomagnification between principal habitats.

Keywords Biomanipulation · Fish · Land use · Stable isotopes · Urban · Wastewater

1 Introduction

Municipal effluents have been identified as a primary cause of increased lake productivity for decades (Hasler, 1947; Rohde, 1969), and remain one of the most significant environmental impacts on lakes (Likun et al., 2017; Smith & Schindler, 2009; Vitousek et al., 1997). As a limiting nutrient in lakes, excessive phosphorus causes eutrophication (Schindler, 1974, 2012), and external loading from anthropogenic origins has been an ongoing concern across the boreal zone. This phenomenon is typically characterised by increased primary productivity, thereby driving the growth of algae and increasing turbidity and hypolimnetic oxygen depletion. Perhaps the most significant change due to eutrophication is that cyanobacteria tend to eventually become the dominant component of phytoplankton blooms (Jeppesen et al., 2005; Schindler, 1977; Smith, 1983). Changes to water quality, as well as changes in abundance and community composition of different species of fish, invertebrates, and macrophytes are typical, and eutrophication may cause fish kills (Schindler et al., 2008; Vollenweider & Kerekes, 1982). Lake eutrophication is commonly a consequence of intensive human settlement, which may reduce the recreational value of these lakes and increase water treatment costs and health risks from cyanobacteria, therefore necessitating monitoring and potential remediation.

Monitoring of lakes in the European Union (EU) is used to classify lake status and identify where remediation may be warranted. Monitoring of water bodies is required under the EU Water Framework Directive (2000/60), including the measurement of neurotoxic mercury (Hg) (European Commission, 2000). Currently in Finland, only one fish species, perch (*Perca fluviatilis*, total length 15–20 cm), is monitored for Hg content (Kangas, 2018). Hg is a global contaminant originating from natural and anthropogenic sources, and the northern hemisphere receives a disproportionately higher amount of Hg atmospheric deposition than is emitted there (AMAP, 2021). Hg

contamination in Europe is changing as emissions from EU countries have continually declined due to improved policy and effluent controls, while emissions from more southerly parts of the world have increased (AMAP, 2021; Dastoor et al., 2022; Lodenius, 2013). However, the burning of fossil fuels and production of cement and non-ferrous metals remain strong contributors within the EU (AMAP, 2021). Additionally, climate change may result in higher levels of Hg in lakes due to increased mobilisation caused by more rain than snow in colder months, shorter winter seasons, and potentially greater atmospheric deposition of Hg (Douglas et al., 2017; Obrist et al., 2018; Pelletier et al., 2021; Steffen et al., 2014). As a contaminant of primarily airborne origin in northern Europe, Hg contamination in lakes is notably complex and modified by the physicochemical and ecological attributes of these ecosystems (Schindler et al., 1995), likely necessitating lake-specific assessments.

Lake Vesijärvi in southern Finland is the 43rd largest lake in Finland (107.5 km²) with a long history of deterioration due to municipal wastewater effluents from the City of Lahti. In the 1970s, sewage water was discharged into the southern Enonselkä basin, resulting in significant eutrophication, dangerous cyanobacterial blooms, and loss of recreational and fisheries value. Toxic cyanobacterial blooms have been associated with such health risks as skin irritation and algal toxin poisoning in humans, as well as fish kills (Igwaran et al., 2024; Lindholm et al., 1989; WHO, 2003). Initially, efforts to improve water quality were taken in the form of improved wastewater treatment and diversion of municipal effluents away from the lake, thereby significantly decreasing the external loading of nutrients (Horppila et al., 1998; Keto & Sammalkorpi, 1988). Despite some level of recovery, within 10 years the Enonselkä basin returned to hosting large blooms of cyanobacteria throughout the year. Additional management and restoration efforts were then made, including hypolimnetic oxygenation in the Enonselkä basin (Keto, 1982) and biomanipulation by intensive trawling for roach (*Rutilus rutilus*) and smelt (*Osmerus eperlanus*) during 1989–1993, resulting in higher abundance of perch and vendace (*Coregonus albula*) (Peltonen et al., 1999). As a result of biomanipulation, fish growth increased (Rask et al., 2020), thereby potentially increasing somatic growth dilution (SGD) of

Hg in tissues (Lavigne et al., 2010; Trudel & Rasmussen, 2006; Ward et al., 2010). At the same time, introduction of pikeperch (*Sander lucioperca*) led to the establishment, and later improvement of the fishery (Ruuhijärvi et al., 2005). However, despite an achieved clear-water state in the 1990s, Enonselkä eventually reverted to poor water quality, cyanobacteria blooms, and deep water hypoxia during the summer (Anttila et al., 2013), which even resulted in a fish kill during the warm summer of 2002 (Ruuhijärvi et al., 2005). Later on, during years of intensive aeration (surface water circulation to deep water in 2010–2018) the smelt stock suffered from the lack of the cool hypolimnetic habitat (Ruuhijärvi et al., 2020). However, hypolimnetic aeration was not found to increase oxygen flux into sediments and high phosphorus releases were still observed, although the periods of hypoxia and anoxia in Enonselkä were shortened (Horppila et al., 2015; Salonen et al., 2020), and the aeration of the hypolimnion was terminated in 2018. This indicates continued nutrient loading beyond the threshold for internal loading from the surrounding Enonselkä catchment (Taipale et al., 2020). Remediating lake eutrophication requires a holistic lake-specific ecosystem approach, and loss of recreational and economic value necessitates such measures for monitoring the changing status of lakes under pressure (Carvalho et al., 2012). While Enonselkä has undergone a history of notable human-impact, the northern Kajaanselkä has not been exposed to as great an extent, and continually maintained higher water quality, less eutrophication, and limited hypoxia in the hypolimnion (KVYY, 2023, 2025). Low elevation, multi-basin lakes are common in the boreal region and may have different morphologies, nutrient profiles, and ecosystem species compositions and abundances (Nelligan et al., 2020; Tammelin & Kauppila, 2018). This is the case with the Enonselkä and Kajaanselkä basins of Lake Vesijärvi (KVYY, 2023, 2025).

To assess trophic hierarchy, stable isotope ratios of nitrogen ($\delta^{15}\text{N}$) are often used due to $\delta^{15}\text{N}$ enrichment with uptake through the food web, which in combination with Hg content of food web organisms can infer Hg biomagnification (Borgå et al., 2012; Cabana & Rasmussen, 1994; Kidd et al., 2012). Increased biodilution of Hg in the food web has been linked to eutrophication due to longer food chain length and increased number of

trophic links and food web complexity, whereby Hg is more broadly distributed throughout the food web (Chen & Folt, 2005; Kozak et al., 2021; Pickhardt et al., 2002; Piro et al., 2025a). Lake eutrophication affects Hg contamination in the food web through food web composition and abundance, as well as energy pathways from benthic to pelagic sources (Eagles-Smith et al., 2008; Hayden et al., 2019). Fish Hg content is typically defined by diet, and biodilution of Hg in more eutrophic environments through the greater biomass of lower trophic levels may result from algal blooms, leading to lower Hg content in fish (Pickhardt et al., 2002; Razavi et al., 2015). In Fennoscandia, low Hg content in fish has been measured in comparatively more eutrophic lakes (Hattula et al., 1978; Johnels et al., 1967). In contrast, the opposite has also been observed and attributed to longer food chain lengths promoting higher levels of contaminants in upper trophic levels (Cabana & Rasmussen, 1996; Kozak et al., 2021). However, eutrophication may also increase the production and transfer efficiency of methylmercury in the food web to the point of surpassing this dilution effect and increasing Hg levels (Fitzgerald & McCarty, 2024). Eutrophication resulting from sewage or agricultural discharge has also been linked to higher $\delta^{15}\text{N}$ (Hansson et al., 1997; Hobson et al., 2012; Ofukany et al., 2014), although lower $\delta^{15}\text{N}$ in the food web has also been linked to fertilisation (Peterson et al., 1993). The direction of fish Hg content and bioaccumulation, and food web Hg biomagnification appear to be inconsistent across eutrophicated lakes, likely depending on the nutrient enrichment source (Ahonen et al., 2018; Kozak et al., 2021; Piro et al., 2025a; Verburg et al., 2014). Eutrophication resulting from forestry operations and peatland ditching has been linked to increased Hg levels, whereas the opposite has been observed in the case of eutrophication from agricultural and sewage inputs (Keva et al., 2022; Kozak et al., 2021; Piro et al., 2025a). While monitoring Hg content in biota and in fish muscle as a human dietary source is required in the EU (European Commission, 2000, 2006), the monitoring of Hg biomagnification is not.

In this study, the food webs of two major basins of large, human-impacted Lake Vesijärvi were sampled to test for potential differences in the Hg dynamics between the food webs of these basins, which have

different levels of productivity. The total Hg content ([THg]) and its bioaccumulation in fish, and the biomagnification of THg in both basins were evaluated. This study assessed two primary questions and predictions.

Question 1. Are there differences in the [THg] and bioaccumulation in the same fish species between basins?

Prediction 1: The [THg] will be higher in all fish species residing in the more productive Enonselkä basin and will exhibit reduced bioaccumulation than those residing in the Kajaanselkä basin, as previously observed along climate-productivity gradients (Ahonen et al., 2018).

Question 2. Does the biomagnification of THg significantly differ between basins?

Prediction 2: The biomagnification of THg will be greater in the less productive Kajaanselkä basin compared to the more productive Enonselkä basin, as detected through significant differences between slopes of the linear regression between [THg] and $\delta^{15}\text{N}$, termed as the trophic magnification slope (TMS), and the intercepts, termed as the [THg] baseline (Kozak et al., 2021).

2 Methods

2.1 Study Area

Lake Vesijärvi is a large boreal lake (107.5 km²; mean depth 6.1 m) (Keva et al., 2022; Piro et al., 2025a) primarily comprising two major basins (Enonselkä and Kajaanselkä) in southern Finland with the city of Lahti located on the southern shore of Enonselkä and the village of Vääksy on the northern shore of Kajaanselkä (61°00' N, 25°30' E) (Table 1; Fig. 1). The average winter (January–March) and summer (June–August) air temperatures are −4.9 °C and 15.8 °C, respectively, and average precipitation is 135 mm in winter and 208 mm in summer (1990–2020, FMI database, 2026). Ice-cover typically forms on the lake in November/December and lasts until April/May (HERTTA database, 2026). The Enonselkä and Kajaanselkä basins are deep (maximum depth of 33 m and 42 m, respectively), and mesotrophic (total phosphorus: 20 µg L^{−1} and 11 µg L^{−1}) with Secchi depths of 2 m.

Table 1 Enonselkä and Kajaanselkä morphology and basic physical and chemical characteristics (average values ± standard deviation measured at 1 m depth during June–August 2022 and 2024 (KVY, 2023, 2025))

Variable	Enonselkä	Kajaanselkä
Basin area (km ²)	26	44
Maximum depth (m)	33	42
Mean depth (m)	6.1	6.8
Temperature (C°)	19.2 ± 2.4	19.1 ± 2.7
Oxygen (mg L ^{−1})	8.7 ± 1.9	7.7 ± 2.9
Oxygen saturation (%)	89.1 ± 19.7	95.0 ± 3.2
Turbidity (FNU)	4.7 ± 0.7	2.0 ± 0.6
Conductivity (mS m ^{−1})	11.6 ± 2.7	10.9 ± 0.3
pH	7.9 ± 0.1	7.8 ± 0.1
Alkalinity (mmol L ^{−1})	0.6 ± 0.1	0.6 ± 0.0
Colour (mg L ^{−1} Pt)	13.4 ± 1.8	10.4 ± 2.1
Total nitrogen (µg L ^{−1})	476.2 ± 60.8	340.0 ± 28.3
Total phosphorus (µg L ^{−1})	19.7 ± 2.2	11.4 ± 2.2
Chlorophyll- <i>a</i> (mg m ^{−3})	13.9 ± 4.8	4.6 ± 1.1
Secchi depth (m)	1.7 ± 0.1	2.1 ± 0.5

2.2 Sampling

Primary producers (benthic algae, filamentous green algae, and macrophytes) were sampled by hand from the shoreline and littoral habitats and manually cleaned at the Lammi Biological Station (LBS) laboratory. Zooplankton were sampled by vertical hauls from the deepest points (Enonselkä: 0–33 m, Kajaanselkä: 0–42 m) in each basin (Table 1) with a zooplankton net (500 µm mesh size, 0.8 m diameter, 3 m length) and stored in plastic containers, while littoral benthic macroinvertebrates were sampled from the shoreline using kick-nets (500 µm mesh size) and handpicking. All samples were stored in cool boxes for transport to the laboratory, where they were identified, sorted, and manually cleaned on the same day as sampling. Zooplankton samples were composited, benthic macroinvertebrates were sorted to genus or family, and all samples were stored in 2 ml plastic tubes.

Fish sampling was conducted in July–August 2022 using Nordic multimesh gillnets (length 30 m, height 1.5 m, comprising 122.5 m wide panels of mesh sizes: 5, 6.25, 8, 10, 12.5, 15.5, 19.5, 24, 29, 35, 43 and 55 mm) following the Finnish standard for Water Framework Directive sampling (described in Appelberg et al., 1995). Briefly, gillnet sampling included

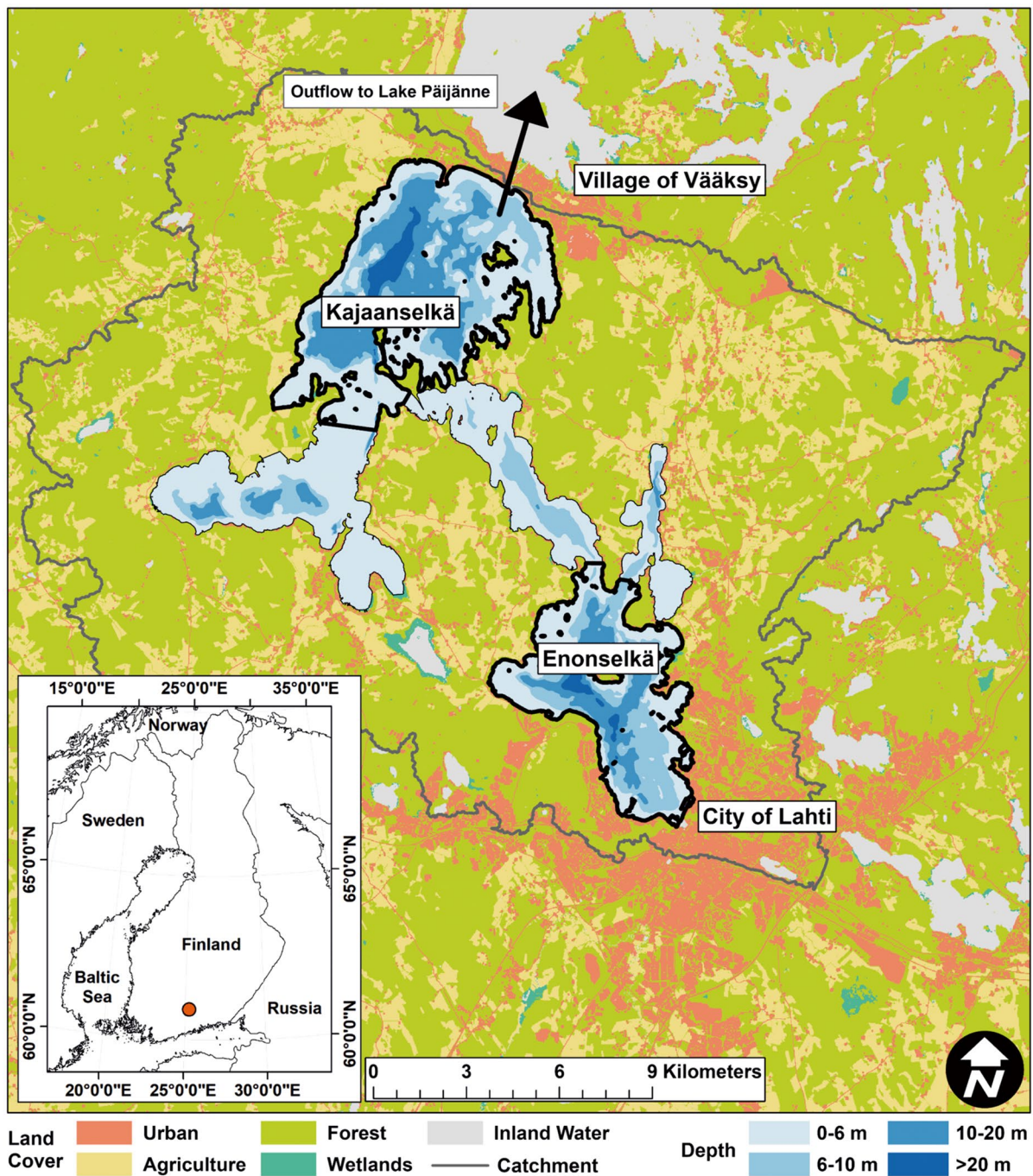


Fig. 1 Lake Vesijärvi and catchment with the City of Lahti and Village of Vääksy locations are indicated, with an arrow identifying the major outflow from Lake Vesijärvi into Lake Päijänne. Kajaanselkä and Enonselkä basins are labelled and

depth profiles as well as catchment land cover are shown. Basin-specific gillnetting locations can be found in Supplementary Figures S1 and S2

stratified random sampling of different depths in littoral, pelagic and profundal zones. Gillnets were set overnight and recovered the following day with a soaking time of 12 h. Fish were removed from nets and euthanised by cerebral concussion for transport on ice to the laboratory at LBS. In addition, samples of species that were not frequently captured in Nordic gillnets (e.g. pike (*Esox lucius*), European whitefish (*Coregonus lavaretus*), vendace, and smelt) were obtained from pelagic trawls (Ruuhijärvi et al., 2020), local fishers, and from a monitoring trap for eel (*Anguilla anguilla*) located at the outflow of Lake Vesijärvi to Lake Päijänne. Fishing licenses were obtained from lake owners as well as a special license from the Centre for Economic Development, Transport and the Environment.

A total of 15 fish species were captured from Lake Vesijärvi, including bleak (*Alburnus alburnus*), bream (*Abramis brama*), roach, rudd (*Scardinius erythrophthalmus*), tench (*Tinca tinca*), white bream (*Blicca bjoerkna*), perch, pikeperch, ruffe (*Gymnocephalus cernua*), vendace, whitefish, burbot (*Lota lota*), eel, pike, and smelt. Fish were identified to species and measured for total length (0.1 cm accuracy) and weight (0.1 g accuracy). A sample of dorsal muscle was taken from 607 individuals to represent the length distribution in total catches (Supplementary Table S1), which were then stored in 2 ml plastic tubes.

Following laboratory work, all plastic tubes were stored in a -20°C freezer. All samples were freeze-dried at -54°C for 48 h using a Christ Freeze Dryer (Alpha 1–2 LDplus) (Martin Christ Gefriertrocknungsanlagen GmbH, Osterode am Harz, Germany) and subsequently pulverised for further analyses.

2.3 Mercury

Individual samples were measured for [THg] using a Direct Mercury Analyzer (DMA-80) (Milestone S.r.l., Sorisole, Italy) (20–30 mg dry weight (d.w.)) at LBS. Blank samples (blank mean $\pm$ SD: $0.00328 \pm 0.0134 \text{ mg kg}^{-1}$, $n=31$) and certified reference materials (DORM-4, National Research Council Canada, mean $\pm$ SD, Hg $0.410 \pm 0.055 \text{ mg kg}^{-1}$ d.w.) (ERM-BB422, European Reference Materials, Hg $0.601 \pm 0.030 \text{ mg kg}^{-1}$ d.w.) were used in the beginning and end of every run (DORM-4:

$0.433 \pm 0.030 \text{ mg kg}^{-1}$ d.w., mean recovery = 102.1%, $n=29$; ERM-BB422: 0.615 mg kg^{-1} d.w., mean recovery = 102.2%, $n=2$). Additionally, where sufficient material was available, samples from 445 of the 607 fish individuals (Enonselkä: 210, Kajaanselkä: 235) were selected with representative length distributions and measured for wet weight (w.w.) [THg]. The same procedure used for dry weight samples was employed in each wet weight run (blank mean $\pm$ SD: $0.00118 \pm 0.00352 \text{ mg kg}^{-1}$, $n=17$; DORM-4: $0.441 \pm 0.029 \text{ mg kg}^{-1}$ d.w., mean recovery = 101.4%, $n=17$). In all sample runs, every fifth sample was duplicated to assess variability.

2.4 Nitrogen Stable Isotopes

Representative subsamples of all freeze-dried materials were weighed and encapsulated in tin cups (animal material: 0.5–0.7 mg; primary producers: 0.9–1.1 mg) for measurement of $\delta^{15}\text{N}$. A total of 231 fish, 47 invertebrate, and ten primary producer samples were measured. Averages were calculated per species in fish, per group in invertebrates, and primary producers were combined (Table 2; Supplementary Table S1). Samples were measured using a Flash EA1112 Elemental Analyzer connected to a Thermo Finnigan DELTA^{plus} Advantage Mass Spectrometer at the University of Jyväskylä. Internal standards with a known reference to Vienna Pee Dee belemnite and atmospheric N_2 were used after every fifth sample, and empty and blank samples were used in the beginning of each run. Analytical error for $\delta^{15}\text{N}$ was $\leq 0.2\text{‰}$.

2.5 Length-adjustment and Bioaccumulation

Comparison of fish [THg] across spatial and temporal scales necessitates an adjustment to reconcile the known covariation between [THg] and fish size (Braaten et al., 2017; Sonesten, 2003). While numerous methods exist attempting to resolve this covariation (Braaten et al., 2017; Knott et al., 2022; Piro et al., 2023), the method from Amundsen et al. (2023) was selected to adjust [THg] for total length (Table 2):

$$_{adj}[\text{THg}] = _{ind}[\text{THg}] \times \left(\frac{_{avg}TL}{TL} \right)^b \quad (1)$$

Table 2 Average total length (TL_{avg}), THg content (dry weight, mg kg^{-1}) (THg), length-adjusted [THg] ($\text{adj}[\text{THg}]$), and $\delta^{15}\text{N}$ (‰) mean $\pm$ standard deviation per species and group used in subsequent analyses with number of samples measured (n) in Enonselkä and Kajaanselkä study basins. Further details are in Supplementary Table S1

Group	Species	Enonselkä					Kajaanselkä						
		n	TL _{avg}	[THg]	adj[THg]	n	δ ¹⁵ N	n	TL _{avg}	[THg]	adj[THg]	n	δ ¹⁵ N
Fish	Bleak <i>Alburnus alburnus</i>	30	13.6±2.6	0.231±0.114	0.218±0.048	10	12.0±5.8	30	13.4±2.7	0.250±0.139	0.240±0.059	10	11.8±5.7
	Bream <i>Abramis brama</i>	28	25.3±9.4	0.178±0.119	0.166±0.075	8	12.7±5.9	27	19.7±7.0	0.134±0.047	0.141±0.037	9	11.9±5.7
	Roach <i>Rutilus rutilus</i>	33	17.0±5.7	0.198±0.124	0.189±0.045	10	12.5±5.8	33	17.1±5.4	0.218±0.114	0.212±0.054	8	11.9±5.2
	Rudd <i>Scardinius erythrophthalmus</i>	11	27.8±4.1	0.339±0.369	0.339±0.369	11	12.0±1.0	8	26.4±5.0	0.419±0.181	0.419±0.181	8	11.0±1.1
	White bream <i>Blicca bjoerkna</i>	32	18.7±6.2	0.363±0.173	0.430±0.162	10	12.9±6.1	16	19.3±5.3	0.212±0.126	0.212±0.126	10	11.5±5.8
	Perch <i>Perca fluviatilis</i>	35	19.7±7.4	0.371±0.219	0.343±0.079	10	14.3±6.6	45	18.5±7.3	0.345±0.204	0.331±0.077	10	13.8±5.8
	Pikeperch <i>Sander lucioperca</i>	31	26.2±11.7	0.230±0.088	0.229±0.049	10	14.9±7.1	28	21.8±11.9	0.297±0.180	0.311±0.064	7	15.1±6.6
	Ruffe <i>Gymnocephalus cernua</i>	30	7.4±0.9	0.097±0.014	0.097±0.014	10	14.5±7.0	30	7.3±1.4	0.162±0.063	0.161±0.047	10	14.4±6.9
	Pike <i>Esox lucius</i>	13	57.9±11.8	0.770±0.451	0.752±0.260	7	14.9±7.7	12	59.4±14.2	0.876±0.577	0.798±0.180	10	14.3±5.6
	Smelt <i>Osmerus eperlanus</i>	37	9.7±0.9	0.123±0.034	0.127±0.026	10	14.4±6.5	30	10.2±1.0	0.248±0.082	0.229±0.059	10	14.0±6.7
Invertebrates	Duck mussel <i>Anodonta</i>	7	0.038±0.008			5	8.2±4.0	21	0.049±0.012			5	7.3±3.2
	Water slater <i>Asellus aquaticus</i>	1	0.038			1	6.5	1	0.040			1	4.4
	Non-biting midge larvae <i>Chironomidae</i>	2	0.025±0.007			1	7.2	1	0.020			1	5.2
	Leech <i>Hirudinea</i>	1	0.020			1	8.7	1	0.014			1	7.4
	Great pond snail <i>Lymnaea</i>	1	0.076			1	6.1	2	0.011±0.001			2	5.6±0.8
	Caddisfly larvae <i>Trichoptera</i>	1	0.017			1	5.6	1	0.016			1	4.4
	Zooplankton <i>Copepoda, Cladocera</i>	22	0.019±0.001			3	9.1±3.2	16	0.016±0.001			3	8.9±3.6
	Combined	4	0.015±0.016			4	5.4±0.7	8	0.005±0.002			8	3.9±1.9

where $_{adj}[THg]$ is the length-adjusted [THg] of the individual, $_{ind}[THg]$ is the measured [THg] of the individual, $_{avg}TL$ is the average fish total length (cm) for the target fish species, and b is the slope of a linear regression between $\log_{10}$ -transformed [THg] and $\log_{10}$ -transformed TL (Supplementary Table S2). The linear regressions were species- and basin-specific. If the linear regression was not significant, then no measurable covariation between [THg] and total length was found and adjustment is not necessary, and measured values were used for basin comparison.

To assess the bioaccumulation of THg in each species, [THg] was $\log_{10}$ -transformed ($\log_{10}[THg]$) to meet assumptions of normality before use in species-specific, basin-specific linear regression with total length (Supplementary Table S3a). The slope of this linear regression is an inference of the bioaccumulate rate, i.e. the rate at which Hg accumulates with length, in each species per basin.

2.6 Biomagnification

Average $\log_{10}[THg]$ was used in linear regression with average $\delta^{15}N$ to infer biomagnification of THg in each basin (Supplementary Table S4a). Biomagnification in Enonselkä and Kajaanselkä basins was assessed using the following equation per basin (Borgå et al., 2012):

$$\log_{10}[THg] = a + b * \delta^{15}N \quad (2)$$

where the estimated intercept of the equation is a , termed as [THg] baseline, and the estimated slope is b , representing the trophic magnification slope (TMS), and $\delta^{15}N$ is the nitrogen isotope ratio ($\delta^{15}N$ $\delta^{14}N^{-1}$).

2.7 Comparative Statistics

The $_{adj}[THg]$ of fish species caught in both basins with equivalent length distributions were tested for significant differences between basins using a t -test. If the assumption of normality using Shapiro–Wilk’s test was violated (in basins where sample size < 30 individuals), differences were assessed using a Mann–Whitney U test, and if homogeneity of variance was violated using Levene’s test, a Welch’s t -test was used (Supplementary Table S5).

Species-specific bioaccumulation was compared between basins with a slope test, with $\log_{10}[THg]$ as the dependent variable and total length as the continuous covariate. An interaction term between basin and total length was included to allow fitting of basin-specific slopes. If the interaction term was not significant, it was removed, and the model was run again to test for significance between intercepts (Supplementary Table S3b). If the linear regression was significant in only one basin, then this was considered a significant difference between basins. The same method was used to test for difference in biomagnification between basins, with $\log_{10}[THg]$ as the dependent variable, $\delta^{15}N$ as the continuous covariate, and an interaction term between basin and $\delta^{15}N$ (Supplementary Table S4b).

The statistical analyses were conducted using R version 4.2.3 (R Core Team, 2019) and significance was determined at $p=0.05$. Figures were generated using the ggplot2 package (Wickham et al., 2007).

3 Results

3.1 THg Content

Of the fish species measured for [THg] between basins, the highest values were observed in pike (Fig. 2; Table 2; Supplementary Table S1). The lowest measurements, however, differed between basins. The lowest in Enonselkä was measured in ruffe, while bream was the lowest in Kajaanselkä. Statistical differences in species $_{adj}[THg]$ between basins were found in white bream, pikeperch, ruffe and smelt ($p<0.001$), and near-significant in roach and rudd (Fig. 2; Supplementary Table S5). With the exception of white bream, fish $_{adj}[THg]$ was higher in Kajaanselkä for those species where a difference was found. Conversion factors for dry weight to wet weight [THg] ranged from 4.58–5.08 in Enonselkä and 2.96–5.11 in Kajaanselkä, and a range of 4.65–5.03 considering species occurring in both basins (Table 3). A conversion factor of 5 is typically used for this purpose (Cresson et al., 2014, 2015; Magalhães et al., 2007), however, the values here demonstrate variability (e.g., eel CF = 2.96).

3.2 Bioaccumulation

Species-specific bioaccumulation slopes were significant for most species in each basin ($p<0.001$),

except for rudd (both basins), ruffe (Enonselkä), and white bream (Kajaanselkä) (Fig. 3, Supplementary Table S3a). Significant differences in slopes were identified in pikeperch and bream, while different intercepts were found in roach, and smelt when slopes were not found to be different (Supplementary Table S3b). In species used for comparison where bioaccumulation slopes were significant, the highest values in both basins were found in smelt, while the lowest were in Enonselkä pikeperch and Kajaanselkä bream. The lowest intercepts in both basins were found in smelt, and the highest were in Enonselkä ruffe and Kajaanselkä pike.

3.3 Biomagnification

Biomagnification of THg was significant in both basins ($p < 0.001$), with a numerically higher TMS and lower [THg] baseline in Kajaanselkä (Fig. 4, Supplementary Table S4a). However, statistical comparison showed that TMS and [THg] baseline were not significantly different between basins. Of the species used to assess biomagnification, the highest average $\delta^{15}\text{N}$ in both basins was identified in pikeperch (Table 2; Supplementary Table S1), although the maximum individual $\delta^{15}\text{N}$ was measured in an Enonselkä pike ($\delta^{15}\text{N} = 17.1$).

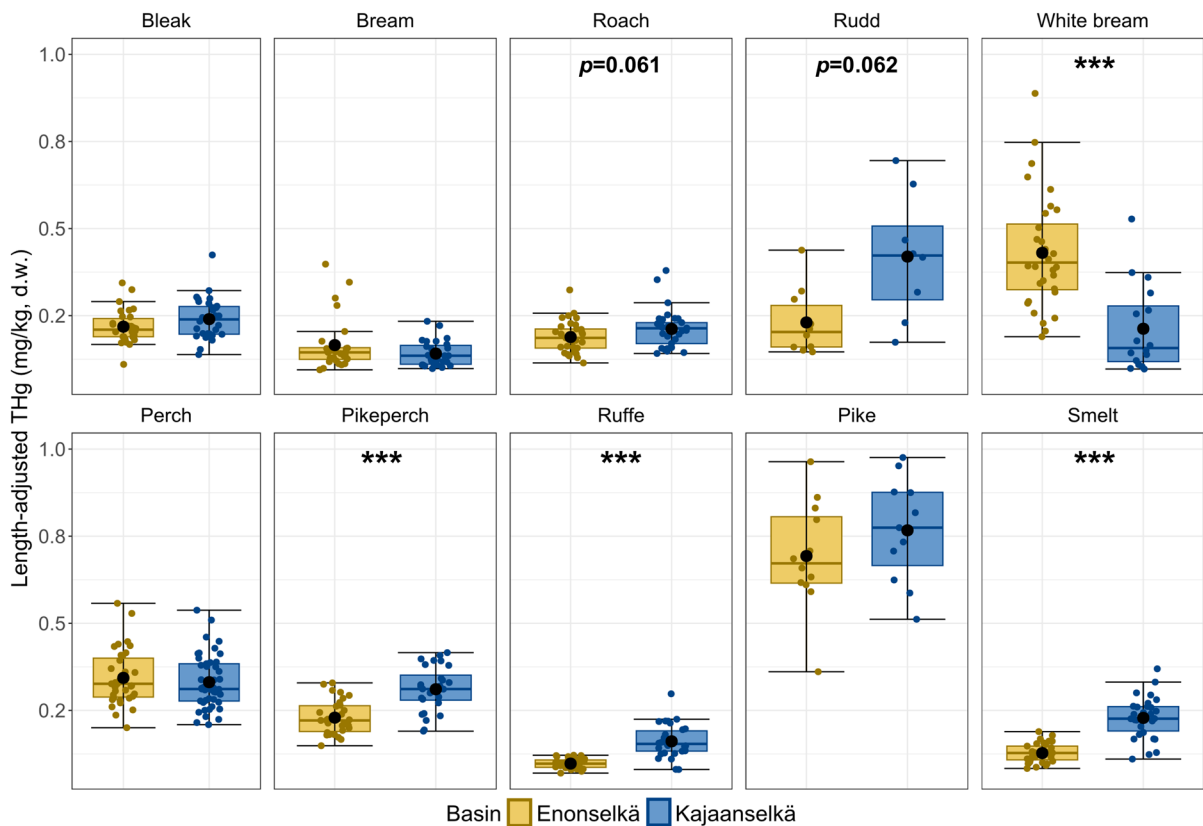


Fig. 2 Basin- and species-specific comparisons of length-adjusted [THg] in Lake Vesijärvi. Data points are horizontally displaced to improve visibility, the black dots in the box plots identify the mean value and the horizontal line identifies the median. Species-specific significant differences

between basins in length-adjusted [THg] are shown with asterisks (***) ($p < 0.001$), and near-significance with p -values (see details in Supplementary Table S5). Outliers have been removed to improve visibility

4 Discussion

4.1 Basin Comparison of [THg] and Bioaccumulation in Fish

Six of the ten species examined were found to have a significant or near-significant difference in [THg] between the basins, and none of the samples exceeded the recommended guidelines for consumption (most species: 0.5 mg kg⁻¹, pike 1.0 mg kg⁻¹ w.w., EU

Regulation 1881/2006) (European Commission, 2006). The [THg] measured throughout the food web was generally lower in most species compared with a series of other Finnish boreal and subarctic lakes with similar nutrient levels (Ahonen et al., 2018; Piro et al., 2023, 2025a). Additionally, bioaccumulation slopes in fish were similar to those in the same species measured elsewhere in Finland, although the intercepts in this study were generally lower (Ahonen et al., 2018; Piro et al., 2023).

Table 3 Species-specific dry weight (d.w.) and wet weight (w.w.) THg content ([THg], mg kg⁻¹ ± standard deviation) per basin and overall with number of samples measured (n) selected for comparison, and w.w.-d.w. conversion factor (CF)

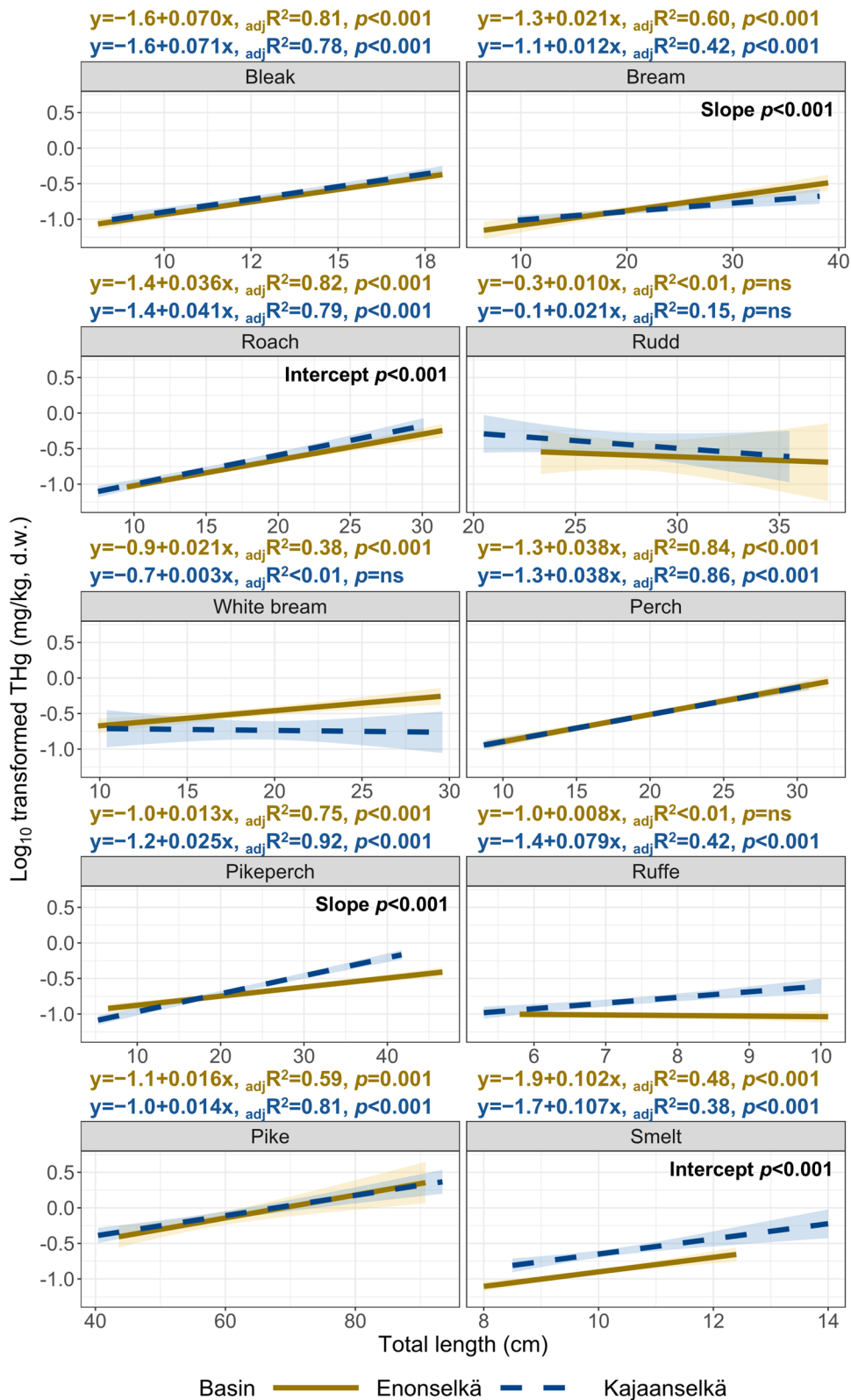
Basin	Species	n	d.w. [THg]	w.w. [THg]	CF
Enonselkä	Bleak	20	0.193 ± 0.068	0.041 ± 0.015	4.73 ± 0.02
	Bream	26	0.180 ± 0.140	0.036 ± 0.024	4.91 ± 0.02
	Roach	26	0.188 ± 0.108	0.039 ± 0.021	4.76 ± 0.02
	Rudd	11	0.339 ± 0.369	0.065 ± 0.053	4.81 ± 0.05
	White bream	23	0.349 ± 0.189	0.069 ± 0.037	5.08 ± 0.04
	Perch	29	0.436 ± 0.315	0.088 ± 0.055	4.82 ± 0.06
	Pikeperch	29	0.228 ± 0.091	0.046 ± 0.019	4.99 ± 0.02
	Ruffe	16	0.098 ± 0.013	0.021 ± 0.004	4.77 ± 0.00
	Pike	7	0.847 ± 0.586	0.173 ± 0.105	4.75 ± 0.11
	Smelt	23	0.110 ± 0.034	0.024 ± 0.005	4.58 ± 0.01
	Total	210	0.259 ± 0.250	0.053 ± 0.046	4.83 ± 0.05
Kajaanselkä	Bleak	20	0.276 ± 0.160	0.055 ± 0.030	4.94 ± 0.03
	Bream	23	0.168 ± 0.131	0.033 ± 0.023	4.94 ± 0.02
	Roach	31	0.208 ± 0.104	0.046 ± 0.023	4.56 ± 0.02
	Rudd	8	0.419 ± 0.181	0.082 ± 0.034	5.10 ± 0.03
	White bream	16	0.212 ± 0.126	0.043 ± 0.025	4.95 ± 0.03
	Perch	27	0.322 ± 0.199	0.065 ± 0.035	4.83 ± 0.04
	Pikeperch	24	0.364 ± 0.141	0.076 ± 0.028	4.77 ± 0.03
	Ruffe	15	0.168 ± 0.053	0.035 ± 0.011	4.84 ± 0.01
	Eel	10	0.410 ± 0.123	0.137 ± 0.033	2.96 ± 0.03
	Pike	12	0.876 ± 0.577	0.169 ± 0.104	5.11 ± 0.10
	Smelt	26	0.250 ± 0.083	0.051 ± 0.015	4.89 ± 0.02
	Vendace	12	0.263 ± 0.117	0.063 ± 0.025	4.10 ± 0.03
	Whitefish	11	0.104 ± 0.041	0.024 ± 0.007	4.20 ± 0.01
	Total	235	0.289 ± 0.237	0.062 ± 0.047	4.70 ± 0.05
Combined basins	Bleak	40	0.235 ± 0.128	0.048 ± 0.025	4.83 ± 0.42
	Bream	49	0.174 ± 0.135	0.035 ± 0.023	4.92 ± 0.42
	Roach	57	0.199 ± 0.105	0.043 ± 0.022	4.65 ± 0.70
	Rudd	19	0.373 ± 0.300	0.072 ± 0.045	4.93 ± 0.59
	White bream	39	0.293 ± 0.178	0.058 ± 0.035	5.03 ± 0.34
	Perch	56	0.381 ± 0.269	0.077 ± 0.047	4.82 ± 0.47
	Pikeperch	53	0.290 ± 0.134	0.06 ± 0.028	4.89 ± 0.39
	Ruffe	31	0.132 ± 0.051	0.028 ± 0.011	4.80 ± 0.40
	Pike	19	0.866 ± 0.564	0.171 ± 0.102	4.97 ± 0.52
	Smelt	49	0.184 ± 0.095	0.038 ± 0.018	4.74 ± 0.62
	Total	412	0.313 ± 0.212	0.063 ± 0.041	4.86 ± 0.11

Higher [THg] in roach, rudd, pikeperch, ruffe, and smelt were found in Kajaanselkä, and higher [THg] in comparatively low productivity lakes and basins has been observed elsewhere in Fennoscandia (Hattula et al., 1978; Johnels et al., 1967; Piro et al., 2025a). This result is generally contrary to the initial prediction of higher [THg] in the more productive Enonselkä basin. Although this phenomenon has been previously linked to comparatively decreased food chain length and biodilution (Kozak et al., 2021), the food chain lengths here are similar between basins. The historical source of increased productivity in Enonselkä is municipal sewage, which has been linked to Hg dilution in fish muscle, whereas lower productivity has been previously associated with increased [THg] due to greater Hg transfer efficiency (Clayden et al., 2013; Hattula et al., 1978; Johnels et al., 1967). Additionally, higher algal productivity in Enonselkä and different feeding strategies of the same species between basins may be drivers of the [THg] difference. The first possibility is supported by recent measurements of lake chlorophyll and phytoplankton biomass, where higher concentrations were measured in Enonselkä (KVY, 2023, 2025). The latter possibility aligns with differing bioaccumulation metrics between basins in the same species.

In species where a difference in [THg] was identified, either a significant difference in bioaccumulation slope or intercept, or a lack of significant bioaccumulation was found. However, the prediction of reduced bioaccumulation in Enonselkä compared with Kajaanselkä was rejected. Lower levels of Hg are typically found in littoral invertebrates compared with their pelagic counterparts (Kahilainen et al., 2017; Power et al., 2002), and fish consuming pelagic resources tend to be more enriched in Hg (Thomas et al., 2016). Consequently, fish relying on pelagic resources have been found to bioaccumulate Hg to a greater extent compared with those relying on littoral resources (Kahilainen et al., 2017). This effect cascades up the food web, as pelagic carnivores feeding on prey with higher levels of THg enrichment increase bioaccumulation metrics (Ahonen et al., 2018), which may be the case in smelt and pikeperch in Kajaanselkä compared with Enonselkä. Additionally, decreased body size and nutritional quality in zooplankton have been observed in Enonselkä,

although this may fluctuate within and across years (Ruuhijärvi et al., 2020; Taipale et al., 2020). In white bream, however, comparatively higher levels of THg measured in Enonselkä individuals are likely linked to the lack of bioaccumulation in Kajaanselkä.

A lack of bioaccumulation in rudd (both basins), white bream (Kajaanselkä), and ruffe (Enonselkä) may result from different habitat use between basins, thereby influencing Hg levels through ontogenetic shifts with consumption of low trophic level food items in littoral areas compared with pelagic areas used in earlier life stages (Hayden et al., 2014; Thomas et al., 2016). Ruffe rely on littoral and profundal resources (Hayden et al., 2015), and the oxygenated profundal of Kajaanselkä facilitate its use to a greater extent than the anoxic profundal of Enonselkä. This may limit resource availability, thus restricting these species to littoral areas in Enonselkä. Profundal resources are typically higher in [THg] than those in littoral or pelagic habitats (Amundsen et al., 2023; Chételat et al., 2008; Kahilainen et al., 2017), which may explain the difference in ruffe Hg metrics. The omnivorous rudd, however, tends to feed on invertebrates and zooplankton at smaller size categories in littoral habitats, followed by increasing herbivory with age (Kapusinski et al., 2015; Nurminen et al., 2003). This likely explains the limited bioaccumulation in this species in both basins. In Kajaanselkä white bream, initial foraging on pelagic zooplankton followed by littoral invertebrates with age (Wielgosz & Tadjewska, 1988; Yazıcıoğlu et al., 2017) likely explains the lack of bioaccumulation. In Enonselkä, bioaccumulation may indicate that white bream mainly persist in the pelagic. This may result from greater competition in littoral habitats due to higher abundance linked to increased productivity (Matern et al., 2021), thus amplifying dietary overlaps with other fish (Specziár et al., 1997; Vinni et al., 2000) and reduced resource availability (Vadeboncoeur et al., 2003). Additionally, pikeperch have been introduced in Enonselkä (Kairesalo et al., 1999), which now has an established naturally reproducing stock. This may limit the use of pelagic areas for some fish as pikeperch tend to prefer feeding on juveniles and smaller, slender species such as roach, perch, and smelt (de Moraes et al., 2025; Keskinen & Marjomäki, 2004; Peltonen et al., 1996).



◀**Fig. 3** Basin- and species-specific bioaccumulation of THg. The linear regression equation, adjusted R^2 , and p -values are given for each species in each basin. Significant slope and intercept comparisons are labelled and represented by p -values (see details in Supplementary Table S3). Note that scales are species-specific

4.2 Biomagnification of THg in the Food Web between Basins

From the broader food web perspective, biomagnification was similar with no significant difference in TMS or [THg] baselines between basins, contrary to the initial prediction. The TMS and [THg] baseline in both basins are similar to lakes of comparable size and productivity previously measured in Canada (Kidd et al., 2012; Lavoie et al., 2013) and Finland (Piro et al., 2025b). Higher TMS has previously been linked to higher nutrient concentration in Canada (Kidd et al., 2012), although the opposite has been observed in a global study (Lavoie et al., 2013) and in subarctic Finland (Kozak et al., 2021; Piro et al., 2025a). Although differences in Hg metrics were noted in the same fish species between basins, these were not apparent when all species were included for assessment of biomagnification. One potential explanation is that when considering the whole food web and the lack of expected difference in biomagnification, these results may indicate recovery of the Enonselkä basin, in conjunction with recent water chemistry measurements (KVYY, 2023, 2025). Extensive efforts to reduce nutrient loads in Enonselkä have taken place, and fewer cyanobacteria blooms coupled with a long-term trend of decreasing phosphorus indicate potential re-oligotrophication (Salonen et al., 2020, 2023). However, comparatively higher methylation potential in the deep, anoxic areas of Enonselkä remains a pressure potentially offsetting biodilution (Fitzgerald & McCarty, 2024; Wang et al., 2023) and equating it with that of the Kajaanselkä basin. This would imply higher [THg] measured in fish from Enonselkä, and the opposite was generally observed. Another potentially confounding factor is the movement of fish between basins (Ofukany et al., 2014). However, based on a tagging study of roach, bream, and pikeperch in Lake Vesijärvi, this seems to have limited importance between Enonselkä and Kajaanselkä (Kotakorpi et al., 2024).

4.3 Eutrophication and Recovery in Lake Vesijärvi, and Monitoring Implications

An extensive history of effluents from the catchment and subsequent efforts towards remediation have produced mixed results. The results of this study in combination with recently measured water chemistry may support re-oligotrophication in Enonselkä (Salonen et al., 2020, 2023). The differences observed here regarding THg bioaccumulation and biomagnification appear limited. This suggests that the baseline Hg and its uptake through the food webs of both basins are generally similar throughout the whole lake, rather than these basins being distinct based on basin-specific levels of inputs seen elsewhere (Hobson et al., 2012; Ofukany et al., 2014). The relationship between phosphorus levels and Hg biomagnification encompasses notable spatial variability (Kidd et al., 2012; Lavoie et al., 2013; Piro et al., 2025a; Verburg et al., 2014), and the rejection of the prediction of higher biomagnification in the less productive basin highlights lake-specific attributes and nutrient sources that must be considered rather than relying on assumptions based solely on trophic status.

This research has been conducted as part of the VesiMon project (<https://www.helsinki.fi/fi/projektit/vesistojen-seurantamenetelmien-kehittamishanke>), which aims to develop and test methods to holistically understand aquatic ecosystems, identify the pressures influencing them, and make monitoring water bodies more cost-effective and accurate. The measurement of Hg in aquatic biota is required under EU Directive 2000/60 (European Commission, 2000), and may provide valuable insight into the source of changes in lakes (Sonesten, 2003), as seen here where generally lower [THg] in fish from the more productive Enonselkä basin align with municipal effluents as a source of eutrophication. While Hg content in perch is typically considered for monitoring purposes in Finland (Kangas, 2018), Hg metrics in these basins were different in pikeperch and smelt. These species were found to have higher [THg] in the less productive basin, coupled with either a higher bioaccumulation intercept or slope than in the more productive basin, whereas no differences were observed in perch. Future monitoring should consider including assessments of similar pelagic species, especially as pikeperch and smelt are abundant pelagic species in

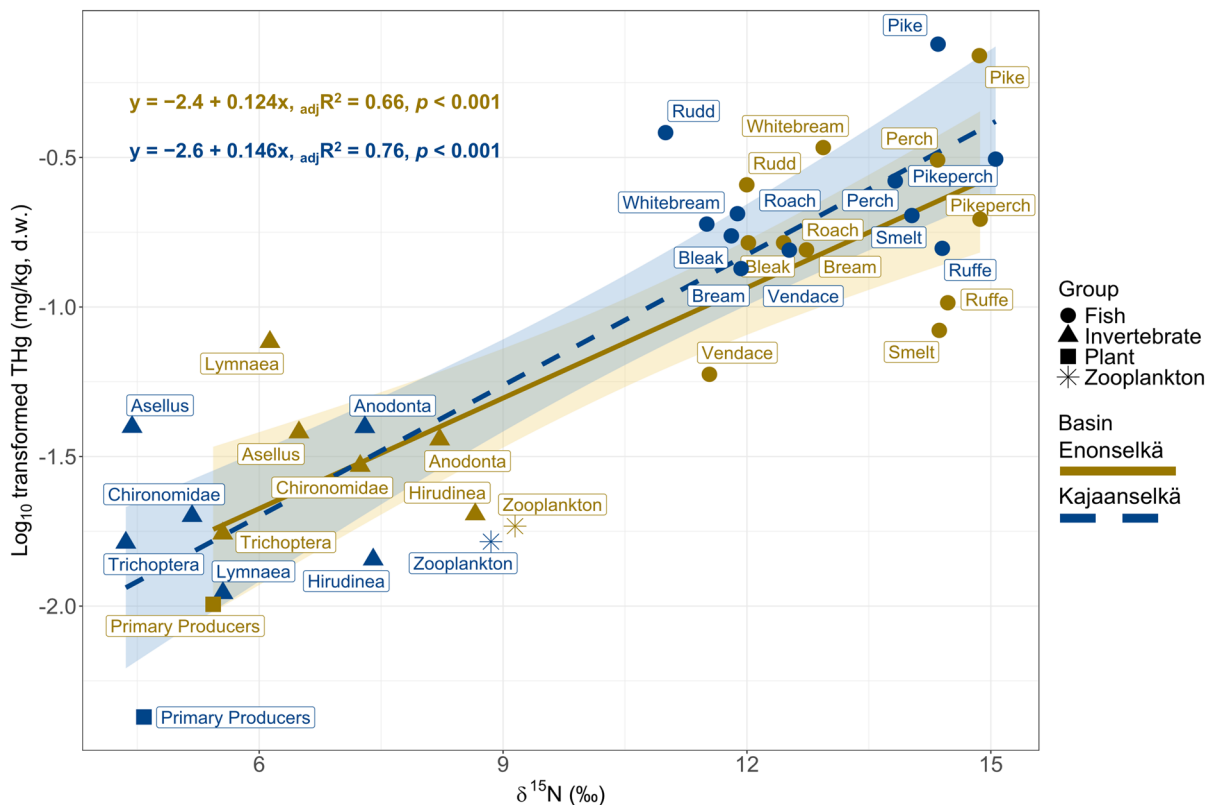


Fig. 4 Basin-specific linear regression of $\log_{10}$ -transformed [THg] and nitrogen isotope values ($\delta^{15}\text{N}$). Group means are indicated by shape, and basin by colour. Primary producers are

basin-specific combinations of algae and macrophyte values. See further details in Supplementary Table S4

many large southern and central Finnish lakes (Jurvelius et al., 2005; Lappalainen & Malinen, 2022), and pelagic prey items may contain higher Hg content (Kahilainen et al., 2017; Karimi et al., 2016). Additionally, monitoring is typically conducted by measuring wet weight [THg] of perch dorsal muscle (Kangas, 2018), to which a conversion factor of 5 may be used to convert from dry weight (Cresson et al., 2014, 2015; Magalhães et al., 2007). Our findings highlight variability within this conversion factor among the species assessed. Combined effects of anthropogenic impacts and their influence on numerous contaminants and transitions of lake status are of significant importance (Schindler et al., 1995). These cause-and-effect relationships necessitate holistic methods due to lake-specific complexity, although appear measurable to an extent in the case of Hg in the food web with value for broader implementation.

4.4 Caveats

The two basins in this study are the major areas of Lake Vesijärvi, however, there are smaller basins in Lake Vesijärvi with higher levels of nutrients and eutrophication, as is the area closer to the city of Lahti. Despite a significant effort in field sampling, ensuring an accurate representation of food webs is a common challenge in such studies. In addition, gaining full length distributions of all fish species present and representative samples of lower food webs remain important pursuits for future efforts (Gozlan et al., 2019; Jackson & Füreder, 2006). Measurement of methylmercury (MeHg) content would have also been of significant benefit, as this is the species of Hg that primarily bioaccumulates in organisms and biomagnifies in food webs. Lower trophic levels typically have lower [MeHg]/[THg] ratios than those higher in

the food web (Arcagni et al., 2018; May et al., 1987), and this ratio increases with age and body size in fish (Lescord et al., 2018). This would imply higher bioaccumulation metrics in fish, and steeper TMS in food webs when [MeHg] is used (Moslemi-Aqdam et al., 2023; van der Velden et al., 2013).

5 Conclusions

This study measured Hg metrics in the food web of two basins in Lake Vesijärvi with different nutrient status and a history of municipal effluents to assess whether Hg could be used to differentiate them. Significantly different [THg] in fish dorsal muscle was observed in half of the species measured, which were typically higher in the less productive Kajaanselkä basin than in the more productive Enonselkä basin. Limited differences were observed in THg bioaccumulation in fish, and in combination with [THg], did not translate to a significant difference in THg biomagnification between these basins. These findings suggest a biodilution effect in the Enonselkä food web, in combination with different species-specific feeding strategies and habitat use between basins. As biomagnification was found to be similar between basins, this may suggest re-oligotrophication of Enonselkä. Measurement of Hg in the food web may be valuable for monitoring purposes in identifying differences between eutrophication sources in lakes and between basins, however, this should be combined with measurements of physical and chemical characteristics and catchment land use analyses. Additionally, as significantly higher [THg] and bioaccumulation metrics were observed in Kajaanselkä pikeperch and smelt, and not in the commonly monitored perch, these or similar pelagic species may be considered for monitoring purposes in large boreal lakes. Pikeperch may be especially valuable for monitoring as this species is commonly consumed as a meal fish. Future studies should assess different basins of large lakes with more pronounced nutrient differences to confirm these findings in a broader context.

Acknowledgements We would like to thank the Lammi Biological Station for providing research facilities and equipment. We thank all individuals and institutions who assisted in the field and laboratory work involved in this study. The illustrations in the graphical abstract were generously contributed by Silja Syvänen. Funding was received from the Finnish Cultural

Foundation as part of the VesiMon Project (JW, TM). KKK thanks the Häme Regional Council, City of Hämeenlinna, and Vanajavesi Foundation for funding.

Author Contributions AJP: Conceptualization, data curation, formal analysis, funding acquisition, investigation, methodology, software, validation, visualization, writing – original draft, writing – review and editing. JW: Resources, writing – review and editing. TM: Data curation, investigation, writing – review and editing. KK: Investigation, resources, writing – review and editing. KKK: Conceptualization, data curation, formal analysis, investigation, methodology, resources, supervision, writing – review and editing, project administration, funding acquisition.

Funding Open Access funding provided by University of Helsinki (including Helsinki University Central Hospital). This work was funded by the Finnish Cultural Foundation, the Häme Regional Council, City of Hämeenlinna, and the Vanajavesi Foundation.

Data Availability Data will be made available upon reasonable request.

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